Name ____ Date ___________ Period _____

Modeling with a Parent Function: the Open Box

TEACHER KEY | Algebra II Honors | Unit 1 | Day 9 | TEKS A2.2.A, A2.7.I

Objective: I can model a situation with a cubic function, justify its contextual domain, locate a maximum numerically, and generalize the model to any rectangular sheet.

The Problem

A flat sheet of cardboard is **20 in. by 10 in.** A square of side **x** is cut from each corner and the flaps are folded up to form an open-top box.

Part 1 — Build and Expand

Each cut removes x from BOTH ends of a side. Write each dimension, then the volume.

height **x** length **$20 - 2x$** width **$10 - 2x$**

$V(x) =$ **$x(20 - 2x)(10 - 2x)$**

Now expand to standard form. Show the multiplication.

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$V(x) =$ **$4x^3 - 60x^2 + 200x$** Degree **3** Family **cubic**

What are the three zeros of the expanded form, and which are physically meaningful?

Zeros at $x =$ **0, 5, 10** meaningful: **none - each gives no box**

Part 2 — Justify the Domain

Why $x > 0$? **a cut cannot be zero or negative**

Why $x < 5$ and not $x < 10$? **the SHORT side runs out first: $10 - 2x$ hits 0 at $x = 5$**

Contextual domain **(0, 5)** Open or closed, and why? **open - both endpoints destroy the box**

Part 3 — Locate the Maximum

Compute V(x). Round to the nearest hundredth where needed.

x	1	2	2.1	2.11	2.2	3	4
V(x)	144	192	192.44	192.45	192.19	168	96

Average rate of change on [1, 2]: **+48** on [2, 3]: **-24**

The sign flips between those intervals. What does that tell you, and why is it stronger evidence than just reading the largest table value?

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The volume stops rising and starts falling, so a maximum lies **between $x = 1$ and $x = 3$**.

Maximum ≈ **192.45 in^3** at $x \approx$ **2.11 in** Box ≈ **15.78 by 5.78 by 2.11**

Range on this domain: **(0, 192.45]** Why is 0 excluded but 192.45 included? **the max is reached; 0 never is**

Part 4 — Generalize

Now let the sheet be **L by W** with W the shorter side, and squares of side x cut from the corners.

V(x) = **x(L - 2x)(W - 2x)** Contextual domain **(0, W/2)**

Explain why the upper bound depends on W and never on L:

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The shorter side reaches zero first, so it **constrains the cut**.

Apply it: a 16 by 8 sheet gives domain **(0, 4)** a 30 by 30 sheet gives domain **(0, 15)**

Part 5 — Going Further

1. For the 16 by 8 sheet, compute V(1) and V(2). Show the arithmetic.

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V(1) = **84 in^3** V(2) = **96 in^3**

2. For a square sheet s by s, write V(x) and its domain. V(x) = **x(s - 2x)^2** domain **(0, s/2)**

3. Does doubling both sheet dimensions double the maximum volume? Predict, then justify.

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No — **volume scales by a factor of 8, since all three dimensions double**

4. A sheet is 18 by 6. Find the domain, then compute V at the midpoint of that domain.

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Domain **(0, 3)** midpoint x = **1.5** V(1.5) = **1.5(15)(3) = 67.5 in^3**

5. Is the maximum always at the midpoint of the domain? Test against the 20 by 10 sheet.

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No — **the midpoint of (0, 5) is 2.5, but the max was near 2.11**

Exit Ticket

A 12 in. by 12 in. sheet has squares of side x cut from each corner.

V(x) = **x(12 - 2x)^2** Domain **(0, 6)** V(2) = **128 in^3**

Explain the upper bound in one sentence: **at x = 6 both sides become 0, leaving no box**